

Noncooperative Game Theory: the Epistemic Approach

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Preface

This is my note on noncooperative game theory. After reading the core of classical game theory textbooks, e.g., Gibbons (1992a) and Osborne and Rubinstein (1994), I find that organizing the theory merely according to solution concepts—like various refinements of Nash Equilibrium—feels LESS natural, as these texts place limited emphasis on explicitly modeling the players' reasoning patterns. I treat game theory as a general model for understanding how players form reasonable beliefs in different settings of interaction and act rationally in the sense that they choose strategies consistent with their beliefs. Suppose the reasoning pattern is clear, i.e., how players form and update their belief is modeled; then the corresponding solution concept can be defined naturally. Indeed, for almost all important solution concepts, one can find a supporting reasoning pattern, and I believe that understanding this underlying reasoning helps clarify each concept's assumptions and limitations.

Initially, I have a vague idea of rearranging the materials in classical game theory textbooks by defining the best-response on beliefs when I first thought about a unified framework of game theory. I discussed my idea with AI and surprisingly found out that epistemic game theory is exactly what I want. My understanding of game theory resembles that described in the introductory part of Brandenburger (2014) and Perea (2012). Besides, I also refer to (copy) classical game theory textbooks (Fudenberg and Tirole (1991), Osborne and Rubinstein (1994), Myerson (1997), Gibbons (1992b) and Gibbons (1992a)) on definitions, results and proofs.

*See [my homepage](#) for updates. I deliberately use "his/her" repeatedly when referring to pronouns, even though this may distract the reader (and myself), for fear of the choice of language raising issues of gender inequality or discrimination. In addition, I use "we" throughout the text to show (and as reminder for me) that that these notes are a shared intellectual journey between the author and the reader.

Abstract

This notes on game theory take an epistemic approach, by which we mean that players are modeled to i) form a reasonable belief and ii) maximize his/her payoff according to the belief. There are multiple reasoning patterns and we follow these patterns to define solution concepts. Instead of treating the epistemic foundation formally, we use the epistemic definitions as intuition and motivation to unfold the classical game theory. This notes have 4 main parts concerning different game theoretic settings, in which we first discuss the setting and then move on to different reasoning patterns and the corresponding solution concepts as well as the refinements. To be specific, each part concerns:

- (1) Strategic Game: We first discuss the complete information case and then move on to Bayesian games.
- (2) Extensive Game with Perfect Information
- (3) Extensive Game with Incomplete Information
- (4) Topics: This part includes some applications of game theory, such as auction, signaling, bargaining...

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Part I

Introduction

In decision theory, we're modeled as acting as if we were maximizing our utility given some exogenous parameters. An example is that producers in competitive markets choose the output that maximizes their profit. In this setting, no individual producer's behavior affects market structure, and expectations about other producers do not influence decision-making.

However, such perspectives of analysis is overly restriction and the real world is not always the case. For example, collusion requires participants to act based on the expectation of other firms' behavior. This is what game theory studies. Game theory is a subject that deals with interaction among decision-makers, focusing on the existence and properties of equilibrium. Equilibrium is a kind of stability, where no participants have incentive to deviate. We'll formalize this concept later on.

1 Game

1.1 Basic Composition of a Game

Now we start to consider the formal definition of games. We try to formalize these concepts by analyzing the game of rock-scissors-pape.

Example 1.1 (Rock-scissors-pape). In this game, we have 2 players and the consequences can be represented in **table 1**, where 1 stands for win, 0 stands for draw and -1 stands for lose. This is also called the payoff matrix, which is widely used in game theory. The embedding assumption of this game is that players prefer winning to losing, inducing a preference relation.

		Player 2		
		R	S	P
Player 1	R	0, 0	1, -1	-1, 1
	S	-1, 1	0, 0	1, -1
	P	-1, 1	1, -1	0, 0

Table 1: Outcome of the Rock-scissors-pape Game

A game consists of players (along with their preferences) and rules (that maps actions to

consequences). Each player i , denoted by A_i , has a set of available actions, called action set $\mathcal{A}_i$. We use a_i to denote a certain action, i.e., $a_i \in \mathcal{A}_i$. As for "rule", it maps the action profile $a = (a_i)_{i \in N}$ to a consequence c , i.e., a mapping g from the Cartesian product of action sets $\mathcal{A} = \prod_{i \in N} \mathcal{A}_i$ to the consequence set C . If we assume the participants are rational, they can surely compare among 2 consequences, inducing a preference $\leq_i^*$ of player i over $C \times C$.

Definition 1.1 (Game). A game is a 4-tuple $\langle N, \mathcal{A}, g, \{\leq_i^*\}_{i \in N} \rangle$, where N is the set of player index. If $\leq_i^*$ admit a utility representation, we may also write $\langle N, \mathcal{A}, g, \{u_i\}_{i \in N} \rangle$ instead.

Notice that the consequence and the action is associated with a mapping, we can equivalently define the preference relation over $\mathcal{A} \times \mathcal{A}$. Let a_{-i} denote the action of other players, i.e., $a_{-i} = (a_j)_{j \in N \setminus \{i\}}$, then the preference of player i can be defined as

$$(a_i^{(1)}, a_{-i}^{(1)}) \leq_i (a_i^{(2)}, a_{-i}^{(2)}) : \iff g(a_i^{(1)}, a_{-i}^{(1)}) \leq_i^* g(a_i^{(2)}, a_{-i}^{(2)}).$$

Then a game can be represented by $\langle N, \mathcal{A}, \{\leq_i\}_{i \in N} \rangle$.

Such a game is of the simplest form since it's almost one-shot and static. We'll introduce history when it comes to the extensive form.

1.2 Strategic Form

A strategic game is a scenario where players make decisions independently and simultaneously.

To understand the meaning of "independently and simultaneously", let's consider an experimental setting, every subject, anonymously sitting behind a computer, is shown a payoff matrix and asked to choose their strategy*. They won't be able to see the outcome until everyone makes their choice. Two important things to understand:

- They don't necessarily to make decisions at exactly the same time.
- It's not necessarily a one-shot game. It can be repeated, but across rounds, no strategic links are allowed.

To make a correct decision, they should care not only about their own choice, but also make decisions under other subjects' possible action, i.e., form a belief and maximize his/her payoff according to this belief. To be more specific, we need to consider two things: how should one interact given opponents' action? And, what are opponents' possible actions? We will formalize this idea soon after we give the definition of a strategic game.

Definition 1.2 (Strategic Game). A strategic game is a 3-tuple $\langle N, \mathcal{A}, \{\leq_i\}_{i \in N} \rangle$. Everything is defined in the introduction part of the book. If every $\mathcal{A}_i$ is finite, we say the game is finite.

* Note that this is the perfect information case. We'll talk about the impact of information in 2.4.

1.3 Extensive Form

1.4 Common Example of Games

We first present some strategic games. The first stream includes some social dilemma games:

Battle of the sexes(BoS) or according to Osborne and Rubinstein (1994), Bach or Stravinsky.

The Prisoners Dillema(PD) No equilibrium as a strategic game, but allows cooperation and defection. It describes a scenario where two suspects are put into two cell and if they both confect..

Hawk-Dove, or Chicken Game Similar to PD, but only one cooperation option.

Table 2: Social Dilemmas

(a) BoS			(b) PD			(c) Chicken		
	B	S		D	C		D	H
B	2, 1	0, 0	D	3, 3	0, 4	D	3, 3	1, 4
S	0, 0	1, 2	C	0, 4	1, 1	H	1, 4	0, 0

2 Belief, Information and Knowledge

2.1 State and Belief

2.2 From State to Information

2.3 Knowledge and Common Knowledge

2.4 Perfect and Incomplete Information

3 Decision Under Uncertainty

3.1 Lottery and the Induced Preference

3.2 von Neumann-Morgenstern Theorem

3.3 Risk

Part II

Strategic Game

Before getting into the definitions, we make clear what our goal is in strategic games as a player. As we assume players are rational, i.e., payoff seekers, we are solving a maximum problem. Given different game settings or solution concepts, there's slight difference in notation or technique. But, generally speaking, a player in a game should form a reasonable belief and make optimal decisions under such a belief.

In general, we consider that a player i with the belief $(\mathcal{A}_{-i}, \pi_i)$, where π_i is a probability measure on $\mathcal{A}_{-i}$, is trying to maximize his/her payoff by choosing a strategy $\sigma_i \in \Delta(\mathcal{A}_i)$, i.e.,

$$\max_{\sigma_i \in \Delta(\mathcal{A}_i)} \sum_{a_i \in \mathcal{A}_i} \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma_i(a_i) \pi_i(a_{-i}) u_i(a_i, a_{-i}), \quad (1)$$

and denote its solution the best response correspondence

$$BR_i(\pi_i) := \arg \max_{\sigma_i \in \Delta(\mathcal{A}_i)} \sum_{a_i \in \mathcal{A}_i} \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma_i(a_i) \pi_i(a_{-i}) u_i(a_i, a_{-i}). \quad (2)$$

As long as the belief is reasonable, a game is trivial as player can do their best by playing a strategy in the best-response correspondence. Thus, we'll first formalize the reasoning process and then turn to solve such a UMP.

For the complete information case, we'll first introduce several settings where belief is determined in different ways, and under such belief, how should players react. In the following section, we introduce the concept of rationalizability, which concerns what belief is reasonable.

4 With Perfect Information

4.1 Rationalizability

4.1.1 Rationality and Strictly Dominated Actions

We now begin to understand the very first reasoning pattern. As a rational player, i.e., a payoff seeker, we should only play strategies in best response because it helps you obtain the highest

payoff given a specific belief. We say such strategies are rational.

Definition 4.1 (A rational strategy). We say a strategy $\sigma_i \in \Delta(\mathcal{A}_i)$ is rational if $\exists \pi_i \in \Delta(\mathcal{A}_{-i}) : \sigma_i \in BR_i(\pi_i)$.

Well, will our opponents choose their strategies in the same way? It's reasonable to assume that they are rational. Thus we assign positive probability only to strategies that can be supported by a certain belief. For actions can not be supported by any belief, we call them strictly dominated actions and they'll not appear in a mixed strategy.

Definition 4.2 (Strictly Dominated action). We say $a_i \in \mathcal{A}_i$ is strictly dominated if $\forall \pi_i \in \Delta(\mathcal{A}_{-i}) : a_i \notin BR_i(\pi_i)$.

We'll see in the next theorem that a strictly dominated action will not be chosen as a pure strategy since there always exists a mixed strategy that yields a higher expected payoff..

Theorem 4.1 (Pearce). a_i is strictly dominated $\iff$ there exists a mixed strategy $\sigma_i \in \Delta(\mathcal{A}_i)$ such that

$$\forall a_{-i} \in \mathcal{A}_{-i} : U_i(\sigma_i, a_{-i}) > u_i(a_i, a_{-i}).$$

Then we say σ_i strictly dominates a_i .

Let NB_i be the set of all strictly dominated actions for player i . Then we can reduce the original game as $\mathcal{G}^1 := \langle N, \prod_{i \in N} (\mathcal{A}_i \setminus NB_i), \{\leq_i^1\}_{i \in N} \rangle$, where $\leq_i^1$ is a restriction of $\leq_i$ on $\mathcal{A}_i \setminus NB_i$.

4.1.2 Hierarchies of Beliefs and Common Knowledge

The idea of rationalizability is that, take a player's view, not only I'm rational, my opponents are also rational. That is to say, any strategy that is not in the best-response correspondence won't be taken. Thus, I should assign probability zero to such strategies in my belief. Also, this is at least mutual knowledge, i.e., everyone in the game knows that every other player's strategy is a best-response.

The above intuition actually mentions 2 kinds of beliefs. One is the belief of a player on his/her own strategies, i.e., positive probability will only be assigned to best-response strategies. The other is the belief on other players' rationality, i.e., whether they assign probability zero to non-best strategies.

4.1.3 Iterated Elimination of Strictly Dominated Actions

The section above shows how a single player eliminates strictly dominated actions, now we extend it to the game level and assume that players have unlimited reasoning.

By the above motivation, we first eliminate the never-best response by reducing the game into $\mathcal{G}^1 = \langle N, \mathcal{S}^1, \{\leq_i^1\}_{i \in N} \rangle$, where

$$\mathcal{S}^1 = \prod_{i \in N} \{a_i \in \mathcal{A}_i : \exists \pi_i \in \Delta(\mathcal{A}_{-i}) \text{ s.t. } a_i \in BR_i(\pi_i)\}, \quad (3)$$

and $\leq_i^1$ is obtained by restricting $\leq_i$ in $\mathcal{S}^1$. (3) says that players won't take actions in never-best response into account, even as a composition of a mixed strategy, thus player should form a reasonable belief that assigns probability zero to other players' never-best response. Denote the set of reasonable belief for player i as $\mathcal{B}_i^1 := \{\pi \in \Delta(\mathcal{A}_{-i}) : \text{supp}(\pi) \subset \mathcal{S}_{-i}^1\}$.

Now we define iteratively, suppose we have defined $\mathcal{G}^1, \dots, \mathcal{G}^k$ and the set of reasonable belief is denoted by $\mathcal{B}_i^k$. For each $k \geq 1$, let $\mathcal{S}_i^k$ be the projection of $\mathcal{S}^k$ onto player i 's action set, and define the restricted best-response correspondence as

$$BR_i^k(\pi_i) := \arg \max_{\sigma_i \in \Delta(\mathcal{S}_i^k)} \sum_{a_i \in \mathcal{A}_i} \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma_i(a_i) \pi_i(a_{-i}) u_i(a_i, a_{-i}).$$

Thus, define

$$\mathcal{S}^{k+1} := \prod_{i \in N} \{a_i \in \mathcal{A}_i : \exists \pi_i \in \mathcal{B}_i^k \text{ s.t. } a_i \in BR_i^k(\pi_i)\}.$$

The player i will update his/her belief as $\mathcal{B}_i^{k+1} = \{\pi \in \Delta(\mathcal{A}_{-i}) : \text{supp}(\pi) \subset \mathcal{S}_{-i}^{k+1}\}$. Finally, by intersecting all $\mathcal{S}^k$, we obtain a set that survives the iterated elimination of strictly dominated actions with belief updates, denoted $\mathcal{S} := \bigcap_{k=1}^{\infty} \mathcal{S}^k$.

4.1.4 Rationalizability

Before giving the formal definition of rationalizability, we provide a result concerning properties of $\mathcal{S}^k$ and $\mathcal{S}$.

Proposition 4.2. *Suppose $\mathcal{S}^k$ is generated as above, then*

- (1) $\{\mathcal{S}^k\}$ decreases monotonically;
- (2) $\exists K \in \mathbb{N} : \mathcal{S}^K = \mathcal{S}^{K+1} = \dots = \mathcal{S}$;
- (3) $\mathcal{S} \neq \emptyset$;
- (4) $\mathcal{S}$ has the best-response property, i.e.,

$$\mathcal{S} = \prod_{i \in N} \{a_i \in \mathcal{A}_i : \exists \pi_i \in \Delta(\mathcal{S}_{-i}) \text{ s.t. } a_i \in BR_i(\pi_i)\}. \quad (4)$$

We follow (4) to formalize the definition of rationalizability.

Definition 4.3 (Rationalizability). For a strategic game $\mathcal{G}$, we say $a_i \in \mathcal{A}_i$ is rationalizable if

there exists $\{S_i\}_{i \in N}$ such that $a_i \in S_i$ and for any $j \in N$,

$$\forall a_j \in S_j : \exists \pi_j \in \Delta(S_{-j}) \text{ s.t. } a_j \in BR_j(\pi_j).$$

4.1.5 Rationalizability and Equilibrium

4.2 Equilibrium

In this section, we make clear, given a reasonable belief, how you should behave.

4.2.1 Pure Strategy Nash Equilibrium

4.2.1.1 Definition There're games that people may stick to one single strategies, see the following payoff matrix. No matter it's player 1 or 2, he/she should only choose action A as

		Player 2	
		A	B
Player 1	A	2, 2	0, 0
	B	0, 0	0, 0

Table 3: A Coordination Game

that's the only strategy that gives them a positive payoff. We refer to this simplification of strategy a pure strategy.

Definition 4.4 (Pure Strategy). A pure strategy is a strategy that assigns probability one to a single action.

We assume that every player is palying a pure strategy, then we can simplify (1) and (2) as

$$\begin{aligned} & \max_{a_i \in \mathcal{A}_i} u_i(a_i, a_{-i}) \\ BR_i(a_{-i}) &= \arg \max_{a_i \in \mathcal{A}_i} u_i(a_i, a_{-i}). \end{aligned}$$

We refer to the steady state where every subject is playing a best-response equilibrium.

Definition 4.5 (Pure Strategy Nash Equilibrium(PSNE)). Denote $BR(a) := \prod_{i \in I} BR_i(a_{-i})$. We say a^* is a PSNE if $a^* \in BR(a^*)$. Or equivalently, a^* is a PSNE if

$$\forall i \in N : (a_i, a_{-i}^*) \leq (a_i^*, a_{-i}^*), \forall a_i \in \mathcal{A}_i.$$

This leads to an important way of finding the equilibrium:

- STEP I: find the best-response correspondence of each player.
- STEP II: find the action profile satisfying $a_i^* \in BR_i(a_{-i}^*)$.

4.2.1.2 Examples BoS; Coordination Game; PD; H-D; Pennis(strictly competitive game)

4.2.1.3 Existence As we've already seen, not every strategic game has a PSNE. In this subsection, we provide some conditions that guarantee the existence. By the definition of PSNE, it's natural to use the correspondence version of fixed point theorem. Here we use Kakutani's fixed point theorem, which says that as long as the correspondence is nonempty, closed and convex, there exists a fixed point.

To suffice this condition, we assume the followings:

- Action sets are nonempty compact convex subset of $\mathbb{R}^n$;
- The preference relation is continuous and quasi-concave on $\mathcal{A}_i$.

Then by Kakutani's fixed point theorem, there exists PSNE.

Theorem 4.3. *For a strategic game $\langle N, \mathcal{A}, \{\leq_i\}_{i \in N} \rangle$, suppose for all $i \in N$, $\mathcal{A}_i$ is nonempty, compact and convex. Further, we suppose $\leq_i$ is continuous and quasi-concave on $\mathcal{A}_i$, then there exists PSNE.*

4.2.2 Mixed Strategy Equilibrium

For many games, sticking with one strategy is somehow stupid. It's like keep playing scissor in the rock-scissor-pape game, which is clearly unwise. So, players may deliberately randomize the strategy played.

Other motivating example concerning the introduction of mixed strategy includes the matching pennis, which is similar to the rock-scissor-pape game, and the relationship between auditors and taxpayers, see Osborne and Rubinstein (1994, p. 37).

The idea above is to give the strategies a probability measure, which introduce the randomness of the strategies being played.

Definition 4.6 (Mixed Strategy). Let $\Delta\mathcal{A}_i$ denote the set of probability distribution on $\mathcal{A}_i$. Then $\alpha_i \in \Delta\mathcal{A}_i$ is a mixed strategy.

4.2.3 Correlated Equilibrium

5 With Incomplete Information

Part III

Extensive Game with Perfect Information

6 Definition and Solution Concepts

6.1 Definition

The main difference between strategic game and extensive game is that strategic games ignore the sequence of play. We'll use history and player function to depict the sequential play and the corresponding strategies. In this part, we focus on extensive game with perfect information. It's perfect information in the sense that the history realized is common knowledge.

6.1.1 History and Sequence of Play

History records, in the game, the action chosen by players sequentially.

Definition 6.1 (History). Let H be a set of sequences whose component is an action, we say H is a set of history if

- (1) $\emptyset \in H$, i.e., initial history is recorded;
- (2) $(a_1, \dots, a_K) \in H \implies \forall k \leq K : (a_1, \dots, a_k) \in H$;
- (3) $\{a_k\}_{k=1}^\infty \in H \implies \forall i \in \mathbb{N} : (a_1, \dots, a_i) \in H$.

We define the following:

Length $l(h)$ is the length of history h , i.e., the length of sequence h .

Terminal Suppose $h_0 = (a_1, \dots, a_K) \in H$, if there doesn't exist $h = (a_1, \dots, a_{K+1})$ such that $h \in H$, then we say h_0 is a terminal. Let H^T denote the set of terminal history.

Player Function This helps us to identify who is going to play. Suppose we have history $h_1 = (a_1, \dots, a_K)$ and $h_2 = (a_1, \dots, a_{K+1})$, then $P(h_1)$ is the player who takes action after h_1 , i.e., $a_{K+1} \in \mathcal{A}_{P(h_1)}$.

Outcome Outcome can be represented by a function $O : H^T \rightarrow \mathbb{R}^N$, which is the payoff every player can get.

Definition 6.2 (Extensive Game with Perfect Information). $\mathcal{G} = \langle N, H, P, \{\leq_i\}_{i \in N} \rangle$ is an extensive game with perfect information where

- N is the set of players;
- H is the set of history;
- P is the player function defined on $H^N := H \setminus H^T$;
- $\leq_i$ is defined on the set of outcomes $\mathcal{O} := \mathcal{O}(H^T)$.

Every realized history is common knowledge to players. If $\max_{h \in H} l(h) < \infty$, then we say the game has finite horizon.

6.1.2 Strategies

Definition 6.3 (Strategy). Define $A(h) := \{a \in \mathcal{A}_{P(h)} : (h, a) \in H\}$, where $h \in H^N$. Player i 's strategy in an extensive game with perfect information is a function $\sigma_i : H^N \cap \{h \in H : P(h) = i\} \rightarrow \mathcal{A}_{P(h)}$, $h \mapsto a \in A(h)$.

Example 6.1 (Strategies in figure 1a). By definition:

Player set $N = \{Parent, Child\}$;

Terminal History $H^T = \{\{G\}, \{B, F\}, \{B, P, G\}, \{B, P, B\}\}$

Outcome

Define the function $\Sigma : H^N \rightarrow H$ such that

$$\Sigma = \begin{cases} \sigma_{Child}, & \text{if } P(h) = Child \\ \sigma_{Parent}, & \text{if } P(h) = Parent \end{cases} ,$$

where $\sigma_{Child}(h) = h \times \{Good, Bad\}$ and $\sigma_{Parent}(h) = h \times \{Forgive, Punish\}$.

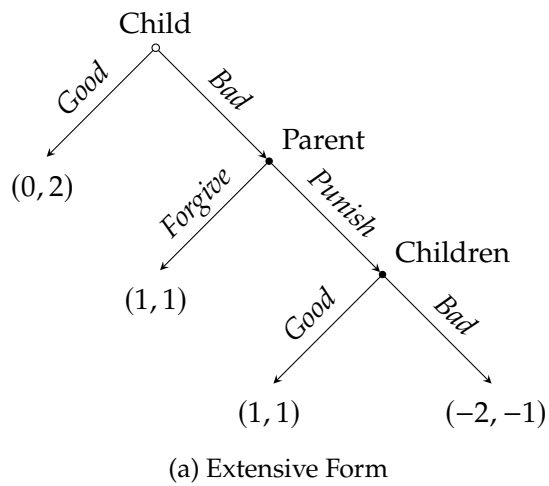
6.1.3 Strategic Form and Reduced Form of an Extensive Game

6.2 Solution Concepts

6.2.1 Nash Equilibrium

This simply treats an extensive game as a strategic game in which the strategies are interpreted as the choice made in a one-shot sense.

Definition 6.4 (Nash Equilibrium(NE)).



	P	F
BB	-2, -1	1, 1
BG	1, 1	1, 1
GG	0, 2	0, 2
GB	0, 2	0, 2

(b) Strategic Form

	P	F
BB	-2, -1	1, 1
BG	1, 1	1, 1
G	0, 2	0, 2

(c) Reduced Form

Note: B = Bad, G = Good, P = Punish, F = Forgive.

Figure 1: Example for extensive form, strategic form and reduced form.

Part IV

Extensive Game with Incomplete Information

Part V

Topics

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References for Part 4

No literature referred.

References for Part 5

No literature referred.